Recovering Versioned Releases Across Build and Publish Boundaries

***Abstract*—An npm package can supply local output to a Docker build, while another image must await registry publication. After partial failure, a consumer can also lose its pending release when its provider's success removes the original propagation reason. We present a protocol combining stage-aware scheduling with consumer-based admission of release intent. Git history determines the plan; tags record completed obligations. The protocol covers single-package repositories, monorepos, and control-repository polyrepos through a common integration contract. We formalize deterministic planning and eight conditional recovery guarantees, with counterexamples to simpler rules. Evaluation of implementation 1.8.2 reports 1,500 agreements in a restricted reference-model domain, 24 recovery episodes, six specification vectors, and eight planning-scale cells. A fresh npm/Docker/Docker experiment checks publication barriers and recovery. Broader-profile disagreements expose oracle limits. The guarantees concern the model; implementation refinement and exactly-once effects at arbitrary external registries remain outside the claims.**

***Index Terms*—release engineering, dependency scheduling, sagas, recovery, idempotence**

I. Introduction

Consider an npm package, a Docker image that packages its output, and another Docker image built from the first. The dependencies cross the boundary between building and publishing. The first image can use files already built in the workspace. The second image, built in an isolated environment, cannot use its new base version until that version is available in a registry. Building all packages before publishing any of them is not a valid schedule for this release. Serializing every package through both stages is sufficient but also forces local-artifact consumers to wait for uploads they do not need.

Interruption makes the problem harder. If the npm package and base image have published but the final image fails, a repeated command should finish the remaining release. It should neither forget that image because its providers have already released nor assign a different version solely because the first attempt made progress. A release engine must therefore answer three questions: which versions are owed, which stages may run, and which obligations remain after a failure. These questions interact, but require different state.

Release engineering spans more than compilation [1]. A heterogeneous release writes to independently administered package registries, container registries, and deployment services. They provide no shared atomic commit. The resulting execution resembles a saga [2]: individual steps commit separately, and recovery must account for the work that already succeeded. For published artifacts, deletion may be unavailable or operationally unacceptable. Our protocol therefore uses forward recovery. It completes or reconciles outstanding work rather than assuming that every external effect has a compensating action.

Workflow engines can express a build/publish DAG. Existing release tools also compute versions and coordinate publication [3], [4], [5], [6], [7]. The question studied here is how to compose stage ordering with a release ledger so that successful providers do not erase their failed consumers' obligations. The key rule is consumer-based admission: a propagated unit of intent remains pending according to the consumer's own release window, even when the originating package has already recorded its release. A second distinction separates intent accumulated over a prerelease train from fresh work not yet delivered.

We present ReleaseWeave, an implementation of this protocol, and RIL, its release-intent language. These names are pseudonyms for anonymous review; reported version numbers identify the evaluated artifacts. The implementation uses Git history and tags as its release ledger and ordinary commands for build and publish stages. It requires no additional persistent release-state database. That claim concerns orchestration state, not the storage used by registries, build tools, or deployed applications.

A package is a versioned release unit: software, specifications, and documentation use the same intent, dependency, command, and completion interfaces. The protocol accommodates different build tools and registries; its guarantees require conforming integrations, not a particular language.

The paper makes three contributions:

- An ecosystem-independent, saga-inspired release protocol combining local-artifact and published-artifact dependencies with explicit release intent and per-package completion records (Section II).
- A self-contained model with deterministic planning and eight recovery, discharge, and scope/channel guarantees under stated assumptions (Sections III–IV). Counterexamples explain why consumer windows and stable retry predicates matter.
- An evaluation of a released binary using differential planning, targeted vectors, injected failures, historical replay, and scale measurements (Section V), with the model-to-implementation gap made explicit.

The mathematical contribution is the admission and discharge discipline, not a new topological-sorting or Git-storage primitive.

II. System and Release Protocol

A. A Release Crossing Two Availability Boundaries

Our running example has three packages: `assets` produces an npm build, `base` packages that output into a Docker image, and `app` builds from the versioned `base` image. Arrows in Fig. 1 run from provider to consumer. The second builder obtains its base through a registry; we do not assume that it shares the first builder's local image store.

An npm label alone does not justify overlap: `base` must actually consume local output. Similarly, Docker does not universally require a push between builds; the registry barrier is needed for the isolated-builder consumption mode in this example. The provider setting `isBuildWaitingPublish: true` declares that its consumers require its published artifact. Here it is set on `base`, while `assets` uses the default local-build mode. Consumer publication is ordered after provider publication in both modes. A missing stage script is a no-op stage, not a reason to omit dependencies or completion records.

Suppose `assets` is at version 1.4.2, `base` at 0.8.2, and `app` at 2.1.0. An intent to release an assets feature and patch all transitive consumers plans versions 1.5.0, 0.8.3, and 2.1.1. After the first two packages publish and are recorded, `app` fails. A planner that only propagates from *currently unreleased providers* can now lose `app`: the original provider is no longer pending, so its intent reaches nobody. The package is an orphan in the release ledger. Our planner instead asks whether `app` has received that intent. Its own pending window is unchanged, so a retry still owes version 2.1.1.

B. From Repository Inputs to a Plan

The planner reads a fixed repository revision, package definitions, dependency edges, commit messages, and release tags. Configuration supplies build and publish commands and availability requirements; manifests can supply package and dependency identities. The plan specifies selected packages, next versions, and reasons before any external publication occurs. Changed inputs require a new plan; a preview is not an authorization for arbitrary later repository states.

Release intent extends Conventional Commits [8]. For example, `feat(assets)^^: add theme assets` directly requests a minor release of `assets` and propagates patch intent to transitive consumers. An unmarked unit affects its own scope only. A scope can explicitly name packages or be derived from changed paths. A commit can contain multiple independent units, each retaining its own source set and propagation depth. The formal model consumes normalized units, not a particular parser syntax. The complete language grammar is outside this paper [9].

Version bumps and prerelease channels are separate axes. A unit can request a channel transition without requesting a bump, and channel resolution can restrict which propagated bumps are meaningful. Cancellation and correction operate on pending intent; they do not unpublish existing artifacts. An exact version directive pins a release, while a hold retains work without admitting the package to the current plan. These controls explain the predicates needed by the model; their CLI forms are not additional scientific contributions.

C. Execution, Recording, and Recovery

For each selected package, the executor schedules version reconciliation, build, and publish work. A consumer's version/build work waits for provider builds, or for provider publications when their availability mode requires it. Build and publish tasks have separate concurrency limits. Independent ready tasks may proceed concurrently; deterministic planning does not require a fixed temporal order for independent completion events.

Successful publication creates a per-package completion obligation. The model records it immediately as a tag. The implementation also has configured recording and finalization steps, whose failures must be reported; their crash boundaries are not assumed atomic with an arbitrary publish command. The next invocation reconstructs outstanding work from the resulting records. This avoids a separate persistent recovery database, but depends on preserving and making the release tags available. A lock prevents cooperating release jobs from writing the same remote release concurrently. It does not make registry operations transactional or prevent an administrator from rewriting history outside the protocol.

Two failure cases must remain distinct. If a publish is known to have failed, its obligation stays pending. If the external operation succeeds but its response or completion record is lost, an absent tag cannot distinguish that success from failure. The integration must reconcile the artifact's identity or use a publisher that safely accepts repetition before recording completion. Forward recovery does not by itself solve this uncertainty [10]. The mathematical claims below refer to the recorded state and make their retry assumptions explicit.

D. Applicability and Practical Limits

The graph may contain one package: the same record-and-retry discipline still applies even though there is no propagation. A monorepo supplies several packages under one history. In a polyrepo, a control repository records submodule pointers and the release graph; its commits and tags provide the single history and ledger used by the model. The linked repositories keep their own histories. This is a construction over a control history, not an atomic transaction across independent Git repositories.

The integration contract has four parts: stable package and version identities; declared dependencies and availability requirements; commands whose successful exit means their promised stage has completed; and durable, trustworthy completion records. Repeated publication must either be safe or reconciled when its outcome is uncertain. These requirements also admit a specification released as versioned files: validation or rendering can be its build stage, and publishing the bundle its publish stage. This is a common protocol for conforming integrations, not an assertion that every existing release workflow or version scheme fits the model. In particular,

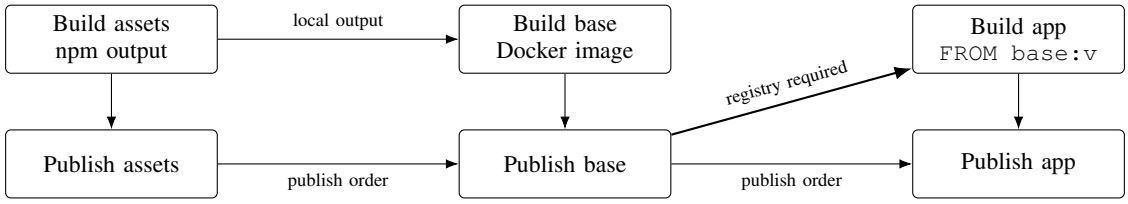


Fig. 1. Stage dependencies, not a measured timeline. Publishing `assets` may overlap building `base`. Building `app` must wait for `base`'s publication. Availability modes are configured requirements: an npm dependency fetched from a registry would also require publication.

the model below assumes an acyclic graph and SemVer-compatible versions.

a) Shared release logic across operating systems: Commands permit different build ecosystems without embedding their build engines in the planner. A game developer can work on Windows while Linux CI builds a dedicated server; conversely, a Linux developer can use Windows CI for native packaging. A shared configuration holds the graph, lifecycle, and portable commands. Small Linux and Windows files reference it and select `sh` or PowerShell, with separate referenced scripts for platform-specific steps. For example, `cmake -build build` can be shared after each platform configures its own build directory. This reuses orchestration logic; it does not translate shell syntax or make build outputs portable.

Conditional and iteration commands can compose shared tasks and gates. However, in 1.8.2 the conditional wrapper itself requires a POSIX shell; a native Windows caller without it must select the platform configuration directly. Toolchains and SDKs must exist on each execution host. These are integration capabilities, not measured Windows/Linux equivalence or evidence of game-development adoption. The experiments evaluate narrower properties.

E. A Stable CI Entry Point

The same package graph can drive named test tasks and the release protocol. A generic CI job invokes those operations rather than enumerating packages and duplicating their dependencies in the workflow. Adding a package within an already provisioned environment changes package configuration and scripts; it need not change the CI workflow. A new runner, credential, permission, or system dependency may still require provisioning changes. Likewise, a configured test gate must explicitly reject missing required tests and block publication on failure: a shared graph alone is not a testing guarantee.

Affected-task execution already appears in systems such as Nx [4]. Here the practical benefit is using one package model for testing and recoverable release execution across artifact kinds. Test and release selection may use different windows and policies; sharing the graph avoids duplicated package lists without replacing CI policy.

III. Formal Model

This section defines the release computation and the publish protocol; Section IV proves guarantees over these definitions. The model separates stage availability from cross-run admission and discharge.

A. Stage Availability

For a selected package p, let B_p and P_p denote successful completion of its build and publish stage, respectively. A version task, when present, is a prefix of B_p. For a provider q and consumer p, the normal successful-stage dependencies include $B_q \to B_p$, $B_p \to P_p$, and $P_q \to P_p$. If q requires published-artifact availability, add $P_q \to B_p$. Here stage arrows mean prerequisite-to-successor; the package edge convention below is consumer-to-provider. The construction is over the full package graph, with stages outside the selected work treated as already satisfied only when their required artifacts are available.

Proposition 1 (stage-order safety)**.** *For a finite acyclic package graph, the constructed stage graph is acyclic. Any schedule that starts a stage only after all its prerequisites succeed respects both local-build and published-artifact availability.*

Proof. Take a provider-first topological rank of packages. Every cross-package stage edge increases that rank. Within a package, the only direction is build to publish. A directed cycle would therefore require a rank decrease or a publish-to-build edge within one package, neither of which exists. Each availability requirement is an explicit prerequisite, so a legal schedule cannot cross it early. The claim concerns ordering, not whether a command actually produces the promised artifact; that is an integration contract. □

A global build phase followed by a global publish phase violates the $P_{\text{base}} \to B_{\text{app}}$ edge in Fig. 1. Conversely, no edge requires P_{assets} to precede B_{base}, so those stages may overlap. This elementary construction uses standard topological scheduling [11], [12]; the recovery results below address the separate problem of retaining the correct selected work.

B. Workspace, State, Windows

Definition 1 (history)**.** The history is a finite DAG of commits with a distinguished $\mathsf{HEAD} = h$; $reach(c)$ is the set of commits reachable from c by parent edges, including c; write $c' \preceq c$ for $c' \in reach(c)$. h is fixed throughout. Normalization supplies a deterministic total order of the commits consistent with ancestry; "newest" refers to this order, with unit position breaking ties within a commit. Ancestry alone is only a partial order, so this ordering policy is a fixed planning input. This reading of a repository is the standard one: the commit DAG with reachability is the working model of git history in

repository mining [13], and the same structure underlies formal derivations of git's three-way merge from least common ancestors [14]. Unlike the patch-theoretic accounts of version control [15], [16], [17], this model tracks no file contents, only history, tags and normalized intent. Manifests and path-derived scopes are supplied through the normalized instance rather than modeled as files.

Definition 2 (workspace)**.** $\mathcal{P}$ is the finite set of packages at h; $E \subseteq \mathcal{P} \times \mathcal{P}$ contains (a, b) when a declares a dependency on b of a configured kind; $dep(p) = \{d : (d, p) \in E\}$. The graph is acyclic; cyclic workspaces are outside this protocol and are rejected before scheduling. For $S \subseteq \mathcal{P}$, $\delta(S, d)$ is the least number of edges on a chain from a member of S to d against the arrows, ∞ if none.

Definition 3 (bumps, versions, channels)**.** $\mathbb{B} = \{\mathsf{none} < \mathsf{patch} < \mathsf{minor} < \mathsf{major}\}$ is a chain, hence a join-semilattice under $\sqcup = \max$. Versions are SemVer 2.0.0 [18] with its precedence; $ch(v)$ is stable for a non-prerelease v and otherwise v's first prerelease identifier; $core(v)$ and $cnt(v)$ are the numeric triple and the prerelease counter; $apply(v, b)$ applies a bump.

Definition 4 (state)**.** A state σ is a finite set of release tags (p, v, c) with $c \preceq h$; states are ordered by inclusion. $base_\sigma(p)$ is the highest-precedence version among p's tags, $sbase_\sigma(p)$ the highest-precedence *non-prerelease* one, $basec_\sigma(p)$ and $scom_\sigma(p)$ their commits; $ch(base_\sigma(p)) = \mathsf{stable}$ by convention when p has no tag. Equal-precedence tags with conflicting release identities are rejected. When no stable tag exists, $sbase$ is the virtual 0.0.0 and $reach(scom) = \emptyset$. An untagged package additionally has $base = 0.0.0$, $reach(basec) = \emptyset$, and a configured initial version; applying its first nonzero bump returns that initial version. Within a valid run sequence, tags are never moved or deleted, so states only grow. Rewriting a completion record is outside the guarantees. Tag write access is therefore release authority; a manifest's version field is not the release ledger. Manifests, configuration, and derived scopes remain trusted planning inputs.

Definition 5 (train and fresh-work windows)**.**

$$W_\sigma(P) \;=\; reach(h) \setminus reach(scom_\sigma(P)), \tag{1}$$

and

$$W^\circ_\sigma(P) = W_\sigma(P) \setminus reach(basec_\sigma(P)), \tag{2}$$

with an empty subtracted reach when P has no tag. W_σ is the *train window*; W°_σ is the *fresh-work window*. When P has no stable baseline, $W_\sigma(P) = reach(h)$: an unreleased package's first release accounts for its whole history. An existing project must therefore configure its initial baseline and scope mapping deliberately. The window is measured from the last *stable* tag so that a prerelease train can recompute its target over the whole train (Eq. 8). Direct admission, propagation admission and ledger discharge use W° instead. Thus a prerelease tag discharges delivered work without erasing the earlier records needed to recompute the train target.

C. Units, Suppression, Admission

Definition 6 (units)**.** $\mathcal{U}$ is the set of units carried by commits in $\bigcup_P W_\sigma(P)$, parsed and with scopes resolved and corrections applied. For $u \in \mathcal{U}$: c_u is its commit, $scope(u) \subseteq \mathcal{P}$ its resolved scope, $bump(u) \in \mathbb{B}$ the bump its type and ! map to; its bump axis is $pb(u) \in \mathbb{B}$ (the stated value, or $bump(u)$ under `inherit`; the default once propagation is asked for is patch because a dependency update need not change the consumer's own API), $pd(u) \in \mathbb{N} \cup \{\infty\}$ (default 0), $ps(u)$ a scope filter; its channel axis is a channel value $pc(u)$, a depth $pcd(u)$ (default 0) and a filter $pcs(u)$.

Definition 7 (suppression and sources)**.** $canc(c, p)$ holds when c is an ancestor-or-self of a `cancel` commit whose scope contains p; $held(p)$ when p's effective `Release-As` is `none`. Both are functions of h and the configuration alone. With $disch_\sigma(p, c) \equiv c \notin W^\circ_\sigma(p)$,

$$src_\sigma(u) = \{p \in scope(u) : disch_\sigma(p, c_u) \vee \neg(canc(c_u, p) \vee held(p))\}. \tag{3}$$

Suppression applies only to *undischarged* work: once p has published the version carrying u, the artefact its consumers are owed is public, and neither a later cancel nor a later hold may retroactively strand them.

Definition 8 (reach)**.** $R(S, \ell) = \{d \in \mathcal{P} \setminus S : \delta(S, d) \leq \ell\}$. Excluding all of S means a unit never propagates to a package it bumps directly; depth is shortest-path and always measured from the originating source set, and a propagated effect never cascades its own propagation, so the depth written in the message is the whole blast radius in every run.

Definition 9 (channels)**.** $chan_\sigma(P)$, resolved for *every* package: the value proposed by the newest eligible direct `Channel` directive naming P with $c_u \in W^\circ_\sigma(P)$ and $\neg canc(c_u, P)$; else the value proposed for P by the newest eligible channel-axis unit u with $pcd(u) > 0$, $P \in R(src_\sigma(u), pcd(u)) \cap pcs(u)$, $c_u \in W^\circ_\sigma(P)$ and $\neg canc(c_u, P)$; else $ch(base_\sigma(P))$. A proposal is computed against $cur = ch(base_\sigma(P))$ (the tag, never a value computed this run) and is empty when it proposes cur itself; a transition proposes only where cur matches its `<from>`; a *propagated* non-transition stable never graduates a prerelease package (graduation must be deliberate); `inherit` resolves to the unit's own `Channel` value if any, else to $ch(base_\sigma(q))$ for its sources q, bytewise least name deciding a disagreement. An empty proposal is ineligible; resolution chooses the newest remaining proposal, with direct proposals taking precedence over propagated ones.

Definition 10 (admission and effective bump)**.** Let $live(u)$ mean $bump(u) \neq \mathsf{none}$, $pb(u) \neq \mathsf{none}$, and $pd(u) > 0$. With $srcCh_\sigma(u)$ equal to $\{chan_\sigma(p) : p \in src_\sigma(u)\}$ and

$$rsv_\sigma(u, D) \equiv \mathsf{stable} \in srcCh_\sigma(u) \vee chan_\sigma(D) \in srcCh_\sigma(u) \tag{4}$$

(the modeled policy admits a prerelease source only on a matching consumer channel; this is an explicit release policy, not a universal claim about package installers),

$$\begin{aligned} adm_\sigma(u, D) \equiv\ & \\ & D \in R(src_\sigma(u), pd(u)) && \text{(reached)} \\ \wedge\ & D \in ps(u) && \text{(in scope)} \\ \wedge\ & c_u \in W^\circ_\sigma(D) && \text{(fresh window)} \\ \wedge\ & \neg\, canc(c_u, D) && \text{(not discarded)} \\ \wedge\ & live(u) && \text{(bump-live)} \\ \wedge\ & rsv_\sigma(u, D) && \text{(resolvable),} \end{aligned} \tag{5}$$

$$eff_\sigma(P) = dir_\sigma(P) \sqcup \bigsqcup\{\, pb(u) : adm_\sigma(u, P)\,\}, \tag{6}$$

where $dir_\sigma(P) = \bigsqcup\{bump(u) : P \in src_\sigma(u) \cap scope(u),\ c_u \in W^\circ_\sigma(P)\}$. Because $\sqcup$ is a join, a package with its own `feat` pending is unaffected by an incoming propagated `patch`, joining an admitted bump cannot lower the effective bump, and gathering order cannot matter.

The consumer-side staleness audit is the upward dual of Eq. 5, not merely reverse reachability. For candidate D it scans units in $W^\circ_\sigma(D)$, excludes D when it is itself a source of that unit, and applies the same shortest depth, scope, cancellation, $live$, and resolved-channel predicate rsv. These clauses make the upward answer exactly equal to the downward planner's obligations, including prerelease workspaces.

D. Plan, Versions, Runs

Definition 11 (plan and versions)**.** Write $pin_\sigma(P) = v$ when the effective surviving exact `Release-As: v` unit for P has $c_u \in W^\circ_\sigma(P)$; otherwise $pin_\sigma(P)$ is undefined. Thus a successful tag discharges a pin just like other fresh work, and the directive is fixed during a retry.

$$\begin{aligned} \Pi(\sigma) = \{\, P : \neg held(P) \wedge (& eff_\sigma(P) \neq \mathsf{none} \\ \vee\ & chan_\sigma(P) \neq ch(base_\sigma(P)) \\ \vee\ & pin_\sigma(P)\downarrow)\,\}, \end{aligned} \tag{7}$$

each member carrying $\nu_\sigma(P)$. Let $S = sbase_\sigma(P)$, $B = base_\sigma(P)$. Define the train aggregate $train_\sigma(P)$ by Eq. 6, replacing its direct and propagated fresh-window tests $c_u \in W^\circ_\sigma(P)$ with $c_u \in W_\sigma(P)$. All other source, scope, suppression, depth, and resolved-channel predicates stay the same. It computes the target bump over the train, while *eff* determines fresh admission. Write $e = train_\sigma(P)$. If $pin_\sigma(P)\downarrow$, it supplies that exact version. A stable package that remains stable uses $apply(S, eff_\sigma(P))$; a prerelease package graduating to stable uses $apply(S, e)$ as required by RIL 2.0 §11.5. Otherwise, with ch the target channel,

$$\begin{aligned} & t = apply(S, e), \\ & \nu_\sigma(P) = \begin{cases} t\text{-}ch.(cnt(B){+}1) & \text{if } core(B) = t \text{ and } ch(B) = ch, \\ t\text{-}ch.0 & \text{otherwise,} \end{cases} \end{aligned} \tag{8}$$

TABLE I
EQ. 8 TRACING ONE TRAIN.

Baseline	Pending bump, channel	Next
`1.2.3`	minor, `beta`	`1.3.0-beta.0`
`1.3.0-beta.0`	no fresh commit	empty plan
`1.3.0-beta.0`	a fresh minor commit	`1.3.0-beta.1`
`1.3.0-beta.1`	minor + a new fix	`1.3.0-beta.2`
`1.3.0-beta.2`	a breaking change lands	`2.0.0-beta.0`
`1.3.0-beta.2`	none, switch to `rc`	`1.3.0-rc.0`
`1.3.0-rc.1`	none, `stable`	`1.3.0`

For a channel-only release whose candidate does not exceed B, recompute once with $e =$ `patch` (the channel-entry patch); an exact pin is never adjusted this way. This permits entry from a clean stable baseline, for example 1.2.0 to 1.2.1-beta.0, without treating an actual bump regression as a channel switch. Every computed or pinned version must exceed $base_\sigma(P)$ in precedence, or the run refuses. Recomputing t from the *stable* baseline every run is what moves a whole train to `2.0.0-beta.0` when a breaking change lands mid-train (Table I).

Definition 12 (run)**.** A run from σ is parameterised by an adversarially chosen set $F \subseteq \Pi(\sigma)$ of packages failing before recorded completion, which induces the blocked set $B = \{p \in \Pi(\sigma) \setminus F : p$ transitively depends on some $q \in F\}$, computed over the *full* workspace graph (a package with no bump this run is still a path from a dependent to a failed dependency). Publication proceeds in a topological order of the full graph, ties broken bytewise by name, filtered to the plan *last*: inducing the graph on the plan first deletes ordering paths that run through unplanned packages. In the abstract transition, each package is tagged after its own publish succeeds, never before (a tag for an unpublished version makes it unrecoverable, since Eq. 1 treats it as released), and *whatever its publish target*, including targets that upload nothing (Proposition 2). The run yields $run(\sigma, F) = \sigma \cup \{(p, \nu_\sigma(p), h) : p \in \Pi(\sigma) \setminus (F \cup B)\}$. The transition abstracts the publish-to-record interval; it does not prove that interval atomic in the implementation. A *run sequence* is $\sigma_1 \leq \sigma_2 \leq \cdots$ with $\sigma_{k+1} = run(\sigma_k, F_k)$.

The propagation bound excludes parsing, history traversal, scope resolution, and output. Those costs depend additionally on reachable commits, parent edges, and retained provenance. Cached traversal may share reachability, but must preserve each unit's source set, depth, filters, channel predicates, and identity. Merging units by directive spelling alone can change admission.

Order matters for a reason tags cannot express: a dependent can be published while its promised provider version is unavailable. If both completion tags later point at the same commit, ancestry alone does not reveal that temporal inversion. The protocol therefore prevents it through stage prerequisites. When a publish is rejected because the version already exists (a run died between publish and tag),

RIL 2.0 requires content-identity verification before adoption and otherwise aborts: tagging a foreign artefact would enrol it into the release lineage. This is a protocol obligation, not a claim that ReleaseWeave can inspect arbitrary user-defined publish scripts; such integrations must supply an idempotency or identity check.

Theorem 1 (determinism)**.** $\Pi(\sigma)$, ν_σ, *the canonical publish order, and the version-guard refusal outcome are functions of the normalized instance I defined below, including its fixed history-ordering policy.*

Proof. Every definition above is a closed expression in those inputs; the three choice points (competing directives, competing propagated channels, publish-order ties) are each resolved by a stated total order: newest commit then last unit within it, and bytewise least name. □

E. Instantiation: What the Theorems Say About a Repository

A normalized instance is

$$I = (\text{history}, h, \mathcal{P}, E, \mathcal{U}, \sigma, \mathit{config}),$$

where configuration includes the fixed history-ordering policy. Repository files and messages supply these objects through normalization. The model abstracts artifact contents, arbitrary hook effects, and other configuration features. Its guarantees quantify over this fragment under acyclicity, version validation, and the run-sequence retry invariant. The implementation is not assumed to check every component of that invariant. Proving that a binary refines the model remains separate work; Section V provides tests of selected behavior, not a refinement proof.

IV. Guarantees and Proof Arguments

Cross-run claims use the *retry invariant*: the corrected tuple stream, source sets, graph, scope/depth predicates, target baselines, and resolved source/target channel admission of every outstanding unit remain unchanged, except for baseline tags written for successful packages. Two recurring components deserve names. **(H1)** says every baseline channel read by `Propagate-Channel: inherit` remains unchanged; **(H2)** says no contribution initially suppressed by cancellation, hold, or correction is later released at the same h. H1 and H2 are necessary in their respective cases but are not, by themselves, the full retry invariant. Neither is needed for one invocation's determinism.

Lemma 1 (window antitonicity)**.** *For $\sigma' = run(\sigma, F)$, $W^\circ_{\sigma'}(p) = \emptyset$ if p published and $W^\circ_{\sigma'}(p) = W^\circ_\sigma(p)$ otherwise. The train window W also empties after a stable release, but may remain unchanged after a prerelease release.*

Proof. The run adds only tags at h. If p gained none, both baseline commits are unchanged. If it published, $basec_{\sigma'}(p) = h$, so Eq. 2 subtracts $reach(h)$ and is empty. Only a stable release also makes $scom_{\sigma'}(p) = h$. □

Lemma 2 (source stability)**.** *Under the retry invariant, $src_{\sigma_j}(u)$ is constant for every outstanding unit throughout a valid run sequence.*

Proof. The corrected tuple stream and its cancellation, hold, correction, and source-set predicates are fixed components of the invariant. H2 is the component excluding later release of an initially suppressed contribution. Arbitrary tag-set inclusion outside a valid run sequence implies no such monotonicity, because a higher-version tag may point to an earlier commit. □

Lemma 3 (retry stability)**.** *Under the retry invariant, an unreleased package's resolved channel, each outstanding unit's source-channel set, and $rsv(u, d)$ remain constant.*

Proof. These are precisely the channel and source-set components fixed by the retry invariant. Definition 9 then selects the same direct or propagated proposal, and Eq. 4 reads the same sets. □

Remark 1. (H1) cannot be replaced by requiring stable completed releases: `inherit` reads $ch(base_\sigma(q))$, which the origin's own graduation moves: with `core` at baseline `1.5.0-beta.0` and commits `feat(core)^++1: x` and `release(core)%stable: graduate`, run 1 plans the consumer at `2.0.1-beta.0` (inherited `beta`) while `core` graduates; if the consumer fails, run 2 plans it at `2.0.1`: same package, both plans, different versions, every completed release stable. Thus unchanged inherited-channel state must be an explicit component of the retry invariant; successful stable releases alone do not imply version stability.

Theorem 2 (the eight guarantees)**.** *Write adm_j for adm_{σ_j} and $n = |\Pi(\sigma_1)|$. Under the retry invariant, for every unit u, consumer D and run $j \geq 1$:*

- G1 ***termination.*** *propagation halts after $O(|\mathcal{U}| \cdot (|\mathcal{P}|+|E|))$ work, whatever the declared depths.*
- G2 ***no orphans.*** *$adm_1(u, D) \Rightarrow adm_j(u, D)$ until D releases at a commit containing c_u.*
- G3 ***version stability.*** *if $D \in \Pi(\sigma_1)$ and D is not released before run k, then $D \in \Pi(\sigma_k)$ and $\nu_k(D) = \nu_1(D)$.*
- G4 ***no double ledger delivery.*** *once D releases past c_u, $adm_j(u, D)$ never holds again; with G2, ledger discharge is exactly-once for every contribution whose target eventually succeeds. This does not make external side effects exactly-once.*
- G5 ***no widening.*** *$\{D : adm_j(u, D)\} \subseteq \{D : adm_1(u, D)\}$.*
- G6 ***convergence.*** *$\Pi(\sigma_{n+1}) = \emptyset$, provided each nonempty retry admits and publishes at least one initially publishable, non-held package.*
- G7 ***channel discharge.*** *a propagated channel contribution is admitted only while its commit remains in $W^\circ(D)$; baseline equality and transition matching independently remove no-op proposals.*
- G8 ***one-way axis independence.*** *bump-axis suppression alone does not suppress a channel change. The*

converse is false: resolved channels are an input to bump admission in Eq. 4.

Proof. G1: the traversal computing R marks each package seen at most once, so it does $O(|\mathcal{P}|+|E|)$ work per unit regardless of depth or cycles; acyclicity is needed for the publish order, not here. G2: compare Eq. 5 clause by clause. Source set, depth, graph, scope, cancellation and channel admission are fixed by the retry invariant. The consumer's fresh window is unchanged until D releases (Lemma 1); that release is the only event that can make its admission clause false. G3: D gained no tag, so its baselines, windows and dir are unchanged; the propagated join is unchanged by G2 ($\geq$) and G5 ($\leq$), whose proofs below use only the lemmas; so eff is unchanged and the channel is unchanged (Lemma 3). Thus bump-only and channel-only membership remain; an exact `Release-As` remains the same because its unit is fixed by the invariant. In every case ν is the same function of unchanged inputs. G4: by Lemma 1, $W^{\circ}(D)$ empties after either a stable or prerelease tag, killing fresh direct and propagated admission. This also makes $pin_{\sigma}(D)$ undefined. This is discharge in the tag ledger; an interruption after an external publish but before its tag requires content-identity reconciliation or safe publisher repetition, not an exactly-once claim about the external operation. G5: every clause of Eq. 5 is constant except the window, which is antitone (Lemma 1); a conjunction of constant and antitone predicates is antitone. G6: $\Pi(\sigma_{j+1}) \subseteq \Pi(\sigma_j)$, since a member of the later plan was not released (else its fresh windows and pins are empty and its channel equals its new baseline) and each disjunct of Π is then unchanged or smaller; by the premise each nonempty retry removes at least one initially publishable package (G4), so $|\Pi|$ strictly decreases to 0 within n runs. The bound assumes progress on every counted nonempty run; it does not bound the number of failed attempts that make no progress. G7: the channel-axis admission test uses $c_u \in W^{\circ}(D)$, so Lemma 1 discharges it. While fresh, equality with the baseline and a transition's `<from>` check separately reject no-ops. G8: the channel pass runs first and reads no bump-axis output; the bump pass then reads its resolved channels through Eq. 4. □

Proposition 2 (tagging every admitted release)**.** *If a released package with no artefact to upload and a non-empty effective bump were versioned but not tagged, its window would never advance (Lemma 1), it would re-enter every plan forever, and G6 would fail. Hence Definition 12 tags whatever is released, and a fully successful model run has an empty subsequent plan. Held packages are excluded from admission. This necessity is relative to a tag-only ledger; another protocol could use other durable completion records.*

Three consequences explain the operational behavior. *Resumption*: after a partial failure, $\Pi(\sigma_2) = F \cup B$ at the original versions (G2–G5), provided the retry invariant holds. *Idempotence*: let $\mathit{release}(s) = \mathit{run}(s, \emptyset)$ denote a fully successful transformation with fixed reviewed inputs and valid records.

TABLE II
RUNNING EXAMPLE AFTER ONE DOWNSTREAM FAILURE. PLANNED VERSIONS ARE ASSETS 1.5.0, BASE 0.8.3, AND APP 2.1.1. THIS IS A MODEL TRACE, NOT A TIMING MEASUREMENT. A NO-PROGRESS ATTEMPT MAY BE INSERTED WITHOUT VIOLATING SAFETY; IT IS EXCLUDED FROM THE PROGRESS PREMISE OF THE CONVERGENCE BOUND.

Run	Plan	Recorded success	Failure
1	assets, base, app	assets, base	app
2	app 2.1.1	app	none
3	empty	none	none

Then

$$\mathit{release}(s) = \mathit{release}(\mathit{release}(s)),$$

under the retry invariant, because the first transformation discharges every admitted obligation and the second has an empty plan (G4–G6). Determinism of an unchanged preview is a separate property (Theorem 1). *Scope preservation*: G5 makes the original plan an upper bound on retry publication under the same invariant. This is not an authorization guarantee against changed commands, rewritten tags, or changed credentials.

The idempotence equation concerns recorded release state. If a publisher succeeds but its completion record is lost, its outcome must be reconciled, or the publisher must safely accept repetition. Arbitrary shell-side effects are not made exactly-once by a tag ledger.

V. EVALUATION

The primary campaign uses the official 1.8.2 ARM64 image and the implemented 2.0 release-intent semantics. Historical competitor measurements remain separate. Our evaluation asks whether planning matches a reference domain, whether recorded progress survives injected failures, and what planning costs on larger graphs. A supplemental stage-order experiment addresses the running example directly; its setup and results are reported below.

A. Study Design and Evidence Boundaries

The reference oracle is approximately 300 lines of Python. It predates the complete formal model and implements a restricted fragment. A seeded generator creates real Git repositories with 5–20 packages, layered DAGs, 4–12 intent units, and mixed release-tag states. Golden cases gate the comparison between the oracle and the released binary. Agreement tests that shared fragment; it does not prove the implementation refines every definition in Section III.

Two profiles answer different questions. The core profile stays within the shared semantic domain. The full profile includes prerelease baselines, channel-only entries, and directives outside the oracle's complete coverage. Full-profile differences are diagnostic observations, not passing conformance checks or automatically established implementation bugs. We retain rejected plans, their diagnostics, and the unit of counting: a history can have several differing package rows.

Recovery campaigns execute interrupted releases and inspect subsequent plans, versions, and tags. Separate fixtures

TABLE III
TWO DIFFERENTIAL PROFILES. DIAGNOSTIC EQUALITY IS NOT EXECUTABLE CONFORMANCE: FOUR MATCHING FULL-PROFILE CASES WERE REFUSED.

Observation	Core	Full
Generated histories	1,500	1,500
Accepted plans	1,500	997
E195 diagnostic refusals	0	503
Matching comparisons	1,500	550
Differing package rows	0	1,914

introduce refused publication and a competing Git push. These observations test selected implementation traces, not every interleaving of external writes. The timing campaign varies graph size, shape, intent-unit count, and unrelated history. It uses a contended host capped at two CPUs and 4 GiB; it is not a controlled cross-tool benchmark. All reported primary counts were reaggregated from saved records for this revision. That integrity check is not a new execution of the original campaign.

B. Planner Agreement and Specification Regressions

All 1,500 core instances agreed, with no invalid sample. Of 1,500 full-profile histories, 997 produced accepted plans and 503 produced complete E195 refusals for proposed regressive versions (Table III). Among accepted plans, 546 comparisons were clean. Four refused dossiers also had no row difference, giving 550 matching diagnostic comparisons overall. Each proposed regression set in the refused dossiers was independently checked by the campaign's validation logic. Rejected diagnostic output is not an executable release plan.

The 1,914 differing rows retain the campaign's heuristic classes: PG1 862, PG2 375, PG2x 14, and PG3? 663, with UNKNOWN 0. These labels partition observations for inspection; UNKNOWN 0 does not show that the oracle explains every current semantic rule correctly. The full-profile results therefore limit the refinement claim rather than extend the 1,500 core agreements to all prerelease behavior.

The binary passed six selected specification vectors, 133–138. They exercise fresh discharge, suppression, channel eligibility and ordering, inherited channel drift, and source exclusion. Vector 135 observes a downward W208 diagnostic, not the upward audit API. These targeted cases complement random core instances but are not complete specification conformance coverage.

C. Recovery and Historical Replay

Two isolated orphan repetitions retained the failed consumer's version and reached an empty third plan. Across two broader recovery campaigns, all 24 episodes converged within four runs, with zero structural G3/G5 check failures. These checks concern the tested schedules and predicates. They do not establish the liveness premise under permanent registry failure, nor do they make an unrecorded external write unambiguous.

Three fresh 1.8.2 fault cells passed 8 orphan checks, 12 clean competing-push checks, and 14 conflicting-push checks.

TABLE IV
SAVED PLANNING MEASUREMENTS. P: PACKAGES; U: INTENT UNITS; C_0: PADDING COMMITS. TIMES ARE SECONDS; RSS IS MAXIMUM MIB. EACH CELL REPORTS THE MEDIAN AND OBSERVED MINIMUM–MAXIMUM, NOT A CONFIDENCE INTERVAL.

P	Shape	U	C_0	Median	Range	RSS
100	layer	200	0	.017	.017–.017	15
1,000	layer	200	0	.250	.235–.252	21
10,000	layer	200	0	19.272	18.992–20.000	92
1,000	chain	200	0	.254	.241–.258	26
10,000	chain	200	0	19.354	18.933–19.645	118
1,000	star	200	0	.244	.237–.247	19
1,000	layer	1,000	0	.290	.280–.302	26
1,000	layer	200	5,000	.664	.640–.665	30

The latter distinguish preservation of completed release tags from incorporation of a concurrent source change. Different scenarios have different assertions; their counts are not a common quality score. The artifact retains the commands, logs, and verdicts needed to see what each assertion establishes.

Five-release historical replay matched 36 of 42 versions in `google-cloud-node` and 7 of 64 in `conventional-changelog`. Those projects did not use the modeled intent language. Their layouts and release policies were reconstructed for the replay, and mismatches may reflect that reconstruction as well as different release conventions. Consequently the experiment demonstrates limits of policy transfer; it is not evidence that our protocol reproduces arbitrary tools' historical version decisions.

D. Comparison with Pinned Release Configurations

The historical refusal and competing-push fixtures pin Lerna 10.0.1, Nx 23.1.2, and Changesets 3.0.1. In those configurations, Lerna and Nx tagged the denied version before its failed upload, while Changesets left it untagged and recovered by querying npm. The observed propagation policies were unconditional for Lerna/Nx, range-gated for Changesets, and per-commit for our protocol. These are observations of the pinned configurations, not universal properties of every mode those tools expose.

In the competing-push fixture, compared runs left tags outside branch ancestry; our historical run merged before its final push and planned the colleague's commit next. The new 1.8.2 fault cells above repeated our side, not the competitors. Release-please's tag timing is characterized from its documented workflow, not an executed comparison [6]. No cross-tool speed or aggregate superiority claim follows from these tests. The comparison isolates ledger and recovery boundaries, which differ from ordinary DAG scheduling.

E. Planning Cost

All eight cells are valid observations under the recorded host conditions (Table IV). Layered graphs have 185, 2,002, and 19,919 edges at 100, 1,000, and 10,000 packages; chain and star graphs have $P - 1$ edges. The 10,000-package costs are substantially above the 1,000-package cells. We do not

infer linear end-to-end scaling from the propagation bound: repository construction, history access, scope resolution, and output work are separate costs. Increasing intent to 1,000 units and adding 5,000 padding commits probe different inputs and should not be merged into a package-count trend. Timing is sensitive to the contended host and changed harnesses, so these observations are not compared with earlier campaign latencies.

The fixtures passed `git fsck`. The optimized `commit-tree` generator matched an ordinary small fixture's commit IDs, references, messages, and log output. This check reduces a fixture-generation threat but does not turn synthetic graphs into a representative sample of industrial repositories. Their purpose is to expose costs and graph shapes explicitly.

F. Counterexamples and Test Obligations

The formalization identifies six failures of simpler rules: reusing the train window for discharge; changing suppression during retries; allowing inherited channels to drift; implementing upward audits with reverse reachability alone; resolving bumps before channels; and merging unit identities during shared traversal. The motivating orphan is a further direct consequence of source-based rather than consumer-based pending work.

These examples explain necessary distinctions in the model. They are not measurements of defect prevalence and are not assertions of outstanding bugs in the released specification. The targeted vectors exercise selected cases; we do not equate the number of logical counterexamples with the number of independently reproduced implementation defects. In particular, the inherited channel example in Remark 1 shows why stable completed versions alone do not establish retry-version stability.

G. Mixed-Chain Availability and Recovery

We also executed the motivating npm $\rightarrow$ Docker $\rightarrow$ Docker chain using the official 1.8.2 macOS ARM64 binary. The npm build produces a file copied into the base image; the app image uses `FROM` with the new base version. Publication uses private Verdaccio and Docker registries. A dedicated BuildKit builder resolves that registry over an isolated Docker network; images are built to OCI archives [19] and published through cached BuildKit exports [20]. No image is published to an external service.

Five configurations ran twice each (Table V). Both availability requirements enabled serialize the two boundaries. Disabling the base's publication requirement makes the app attempt to resolve an unpublished image. The other configurations inject either a base-publication refusal before its push or an app-build failure. Clearing the injected failure and running the same release command tests recovery, without changing release intent. Fixed three-second npm and two-second base-publication delays make ordering observable; these are experimental controls, not workload latencies.

In the mixed runs, base building overlapped npm publication; in the serialized runs it began afterward. App building followed successful base publication whenever that barrier was enabled. Removing it produced a registry "not found" error in both repetitions. All four repaired failures retained the planned versions and completed. All eight successfully completed configurations then reran without another build or publication. The verifier checks 106 assertions over stage events, outcomes, versions, tags, and cleanup; these assertions are not independent experimental samples.

TABLE V
FRESH MIXED-CHAIN EXPERIMENT; TWO RUNS PER CONFIGURATION. FAILURES IN THE FINAL THREE ROWS ARE INTENTIONAL. THE BARRIER-REMOVED CONFIGURATION IS NOT REPAIRED AND IS EXCLUDED FROM UNCHANGED-RERUN CHECKS.

Configuration	First outcome	Retry after repair
Mixed availability	2 successes	not needed
Both barriers	2 successes	not needed
Base barrier removed	2 app failures	not attempted
Base publish fails	2 blocked apps	2 successes
App build fails	2 app failures	2 successes

Event ordering and elapsed intervals use Python 3.12's shared monotonic clock [21]. Earlier setup attempts are excluded: a host-loopback registry was unreachable from the Docker daemon, and Python 3.9 on this host supplied process-relative clock origins unsuitable for cross-process intervals. The retained experiment supports stage ordering and selected recovery paths, not distributed-clock accuracy, production registry reliability, or comparative throughput.

VI. Discussion and Threats to Validity

a) Meaning of the guarantees.: Safety and eventual progress are different obligations [22]. Our retry theorem fixes corrected intent, source membership, admission predicates, and relevant channel inputs. Some stability lemmas are direct consequences of that invariant. Their operational value is to expose the conditions that a retry must preserve, not to claim that any repository mutation is harmless. New commits, edits to release policy, or tag rewrites can require a new plan. A process that fails forever need not converge.

The model abstracts external artifact contents and treats successful publication followed by a completion record as one transition. Lost responses and failed recording remain integration obligations. A Git lock coordinates cooperating jobs but is not a transaction manager for arbitrary registries. The implementation's own-reason failure exceptions, recording stages, and finalization behavior require more detailed refinement than the abstract blocked-set rule. We do not claim that testing selected traces closes that gap.

b) Internal and construct validity.: The author developed the system, model, oracle, and tests, so common mistakes can survive their agreement. The restricted oracle does not cover all current semantics; full-profile categories are heuristic. A passing integrity check shows consistency of supplied records, not independent replication. The concurrent-push cases cover selected interleavings, not a systematic sweep. The measured counts use different units: histories, package rows, episodes, and assertions must not be pooled into one success percentage.

c) External validity.: Synthetic repositories and local fault fixtures establish behavior under controlled conditions. The historical subjects require reconstructed policies, and the comparison uses pinned tool configurations. No user study establishes that authors write correct intent or that teams find it easier to review. Polyrepo applicability follows from the control-repository construction; its operational scale is not measured. Cross-platform scripts, agent gates, and multiple ecosystems demonstrate integration scope, not general effectiveness. Performance observations from one contended host are not service-level promises.

VII. Related Work

a) Release and build systems.: Adams and McIntosh identify release engineering as a distinct research problem [1]. Large monorepo practice [23] and build-system models [12] address dependency ordering and reuse. Incremental build work also formalizes dynamic dependencies [24]; our model instead uses a fixed release graph and records versioned publication obligations. We claim neither incremental rebuilding nor a novel topological algorithm. Lerna, Nx, Changesets, release-please, and semantic-release provide different versioning and publication workflows [3], [4], [5], [6], [7]. Section V-D compares operational traces of specified configurations, not formal reductions of those complete tools. Spinnaker already supports parallel pipeline stages and execution histories [25]; expressing a DAG is not unique to this protocol. Nix identifies component instances through input-derived store paths [26], while TOSCA standardizes service templates and lifecycle interfaces [27]. OpenGitOps calls for versioned desired state and continuous reconciliation [28]. Our ledger instead tracks completed versioned releases across independent publication targets. We neither replace these systems' state models nor claim their guarantees for arbitrary scripts.

Version intent is not a proof of API compatibility. Studies of versioning and ecosystem change [29], [30], [31] motivate explicit policy but do not show that a declared bump is correct. Package-installability work such as EDOS [32] models a different consistency question: a set of installable packages rather than the remaining obligations of an interrupted publication. Formal VCS work studies contents, patches, and merges [15], [16], [17], [14]; Git-mining work explains why history and reference interpretation need care [13]. Our ledger uses those repository objects while leaving content and merge correctness outside its proofs.

b) Recovery and formal assurance.: Platform-independent deployment models already formalize lifecycle and version compatibility: Liu and Smith's buildbox includes build, install, ship, and update transitions [33]. Aeolus studies component states, dependencies, and configuration reachability [34]. These are direct antecedents to a common lifecycle abstraction. Our question is narrower: preserving versioned publication obligations across partial executions. The NEPHELE orchestration report also explicitly uses saga patterns for distributed lifecycle operations [35]; applying sagas to operations is therefore not itself a novelty claim. Sagas [2] and the limits of transaction abstractions [36] provide the recovery setting. Independently committed and idempotently processable operations [37], [10] explain why uncertain outcomes need explicit treatment. Our contribution specializes admission and discharge to versioned consumers; it is not the invention of forward recovery. Anvil [38] verifies reconciliation properties of Kubernetes controller implementations, whereas our handwritten proofs establish a release model and our tests sample implementation behavior. This is a significant difference in assurance. Methodological cautions about fuzz evaluation [39] likewise motivate separating core agreement from unsupported generalization.

VIII. Conclusion

A release graph can cross build and publish boundaries. Local-artifact consumers may proceed after a provider builds; registry-artifact consumers must wait for publication. Correct ordering alone does not ensure recovery: a failed consumer must retain its release obligation after successful providers leave the plan. Consumer-based admission and separate train and fresh-work windows make that obligation explicit. Under the stated retry invariant, the model preserves versions, discharges recorded work, and converges when attempts make progress. Released-binary experiments support selected behaviors while exposing limits in oracle coverage and historical policy replay. Mechanized proofs and a fuller implementation refinement, including recording failures and randomized interleavings, remain open work [40].

Data Availability

A curated local review bundle contains harness code, selected raw records, per-file checksums, and an offline verifier. It reaggregates reported campaign counts but omits binaries, complete Git histories, and some large dossiers; it is not a complete executable replication package. The supplemental mixed-chain experiment has separate code, event logs, and verdicts. Anonymous distribution is being prepared. Identifying source and artifact links are withheld in this working review draft; no public artifact availability is claimed. An archival replication package is planned for publication.

AI Assistance Disclosure

Most AI-assisted work used Claude Fable 5, with additional assistance from Claude Opus 5, Claude Fable 5.1, Gemini, and OpenAI Codex. Across these tools, assistance covered exposition, proof arguments, harness code, experiment execution, and results summaries; it extended beyond language editing. Measurements come from program executions, not language-model predictions. The human author is responsible for the claims, proofs, citations, and interpretation.

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